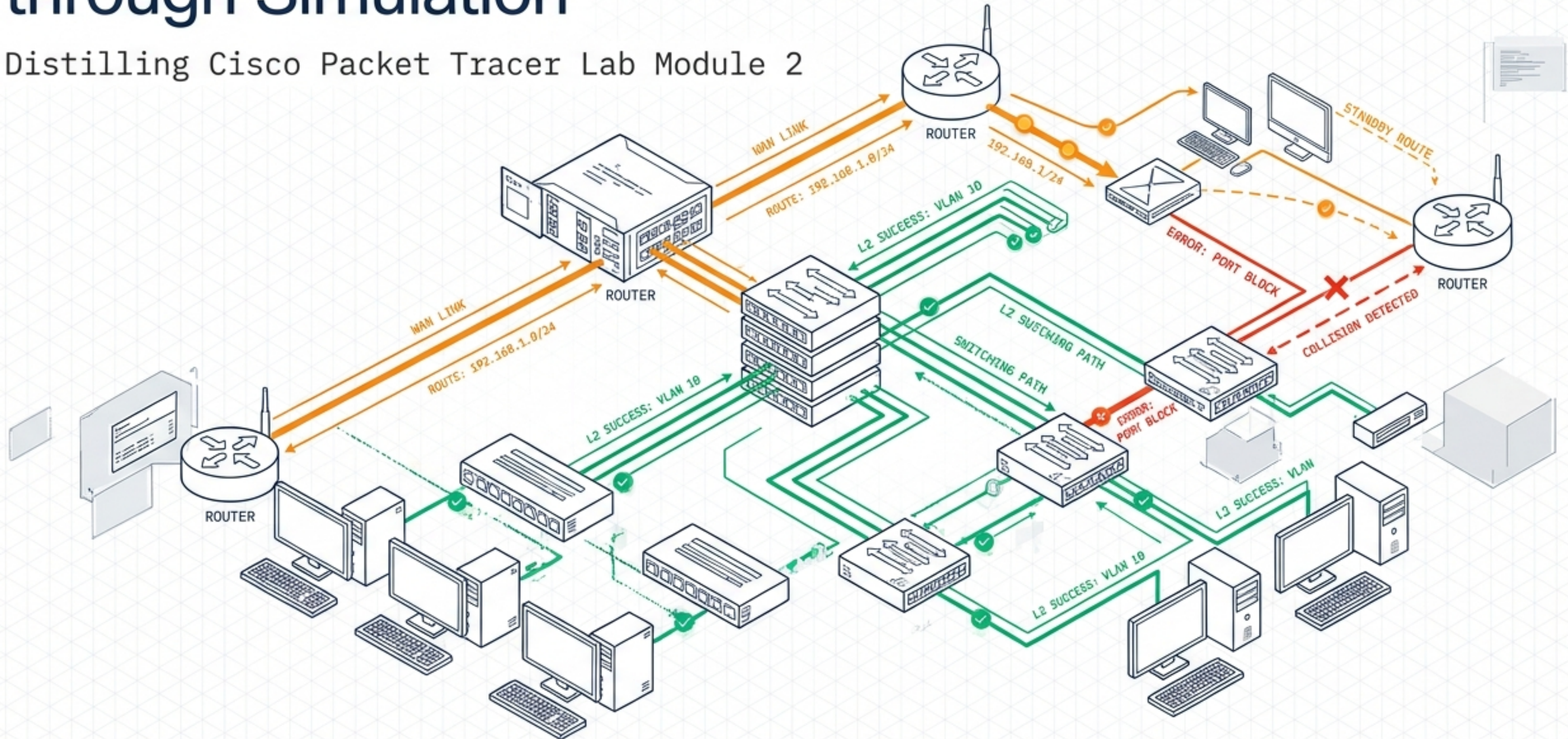


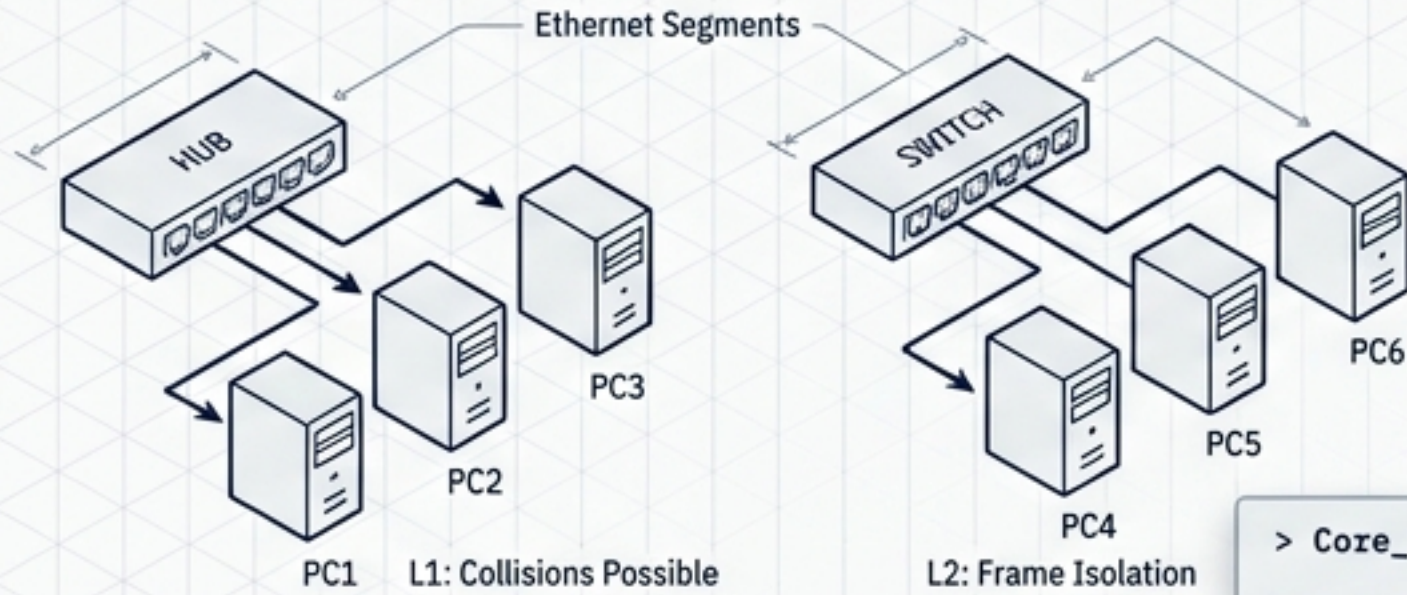
The Physical World: Proving Network Theory through Simulation

Distilling Cisco Packet Tracer Lab Module 2

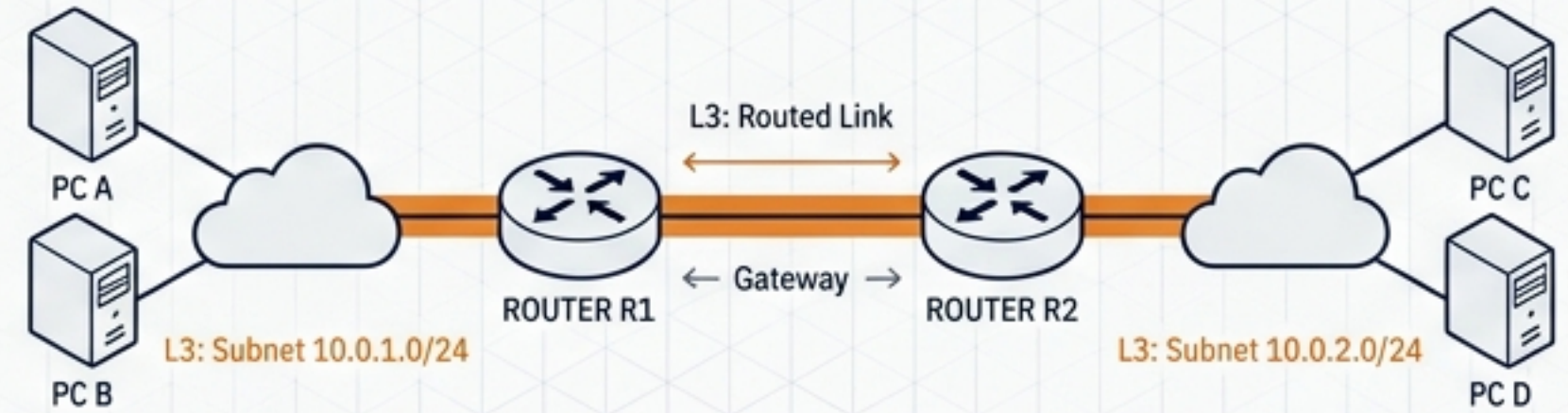


The Lab Architecture

Topology A: Forwarding & Collisions



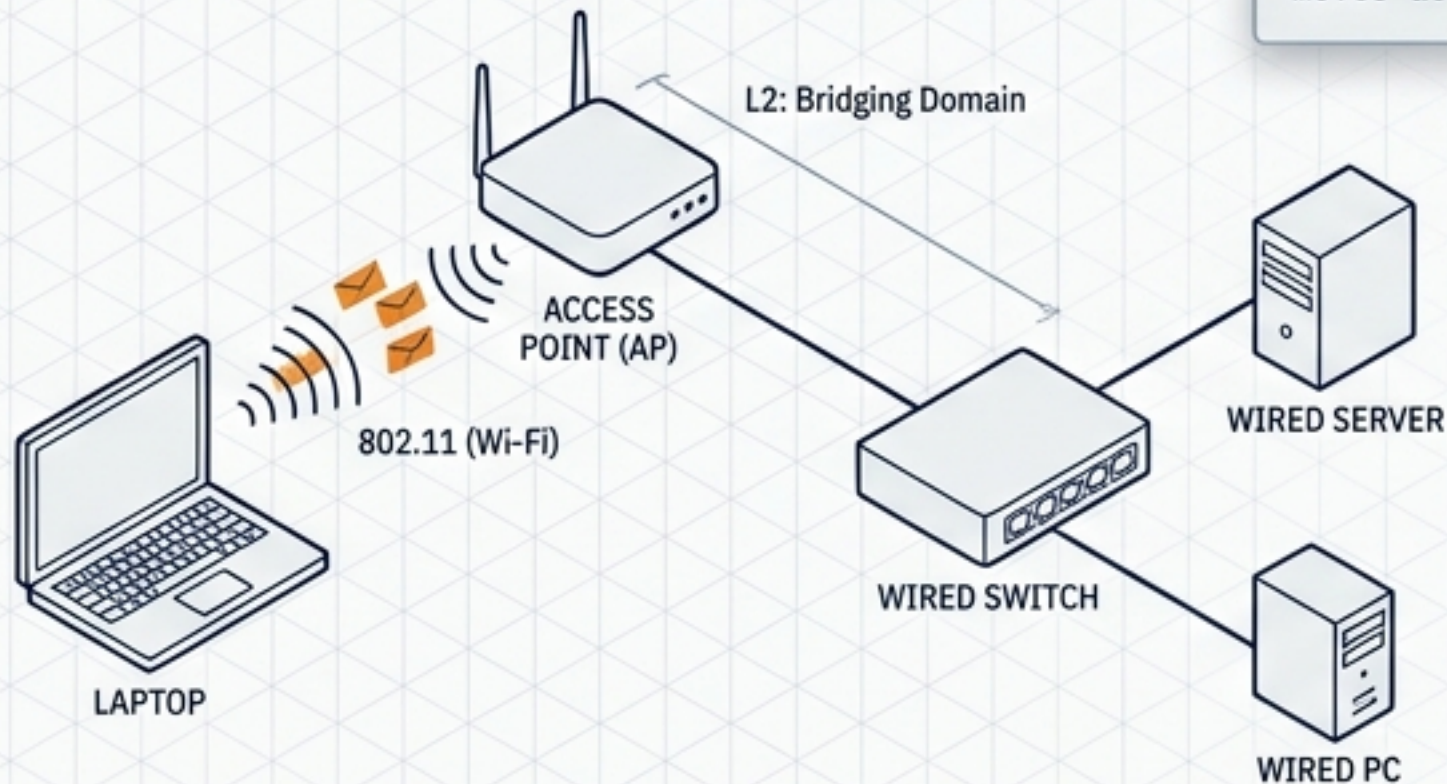
Topology B: Routing & Boundaries



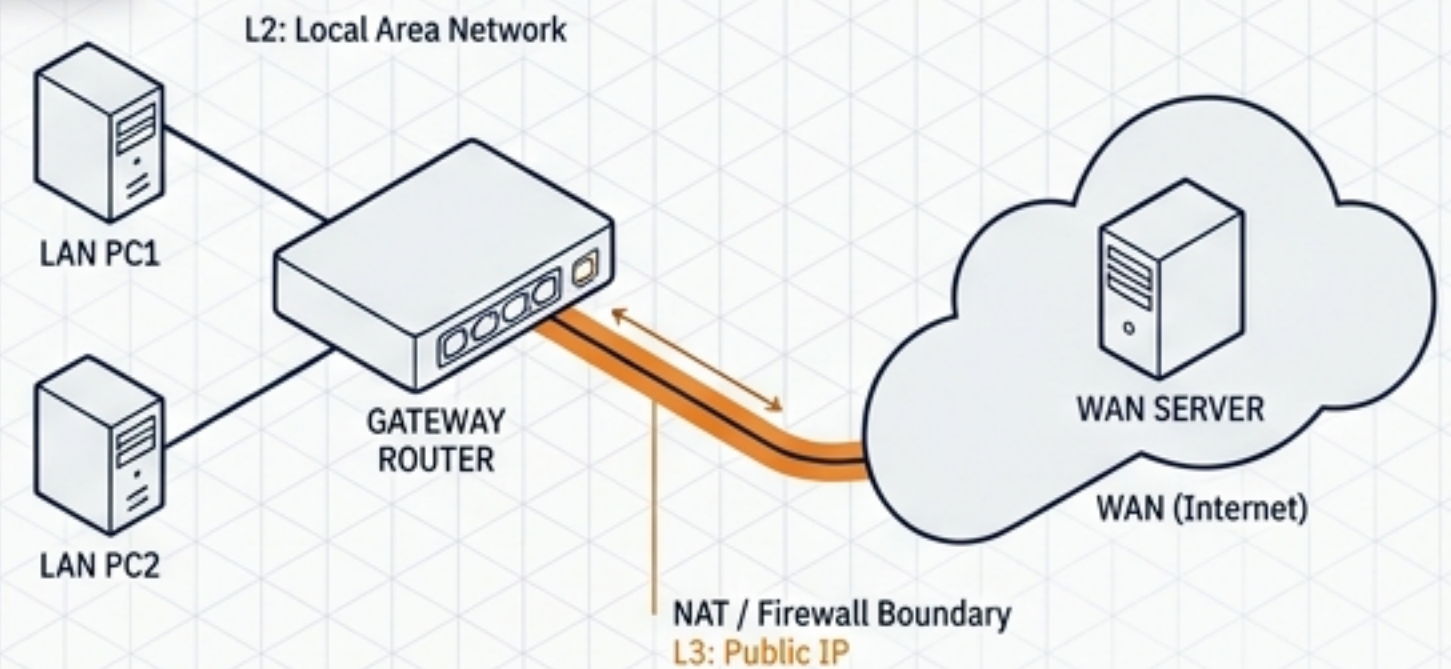
> Core_Premise.exe

Every networking concept is either directly observable or logically provable. This deck isolates each OSI layer to prove exactly how data moves across the wire.

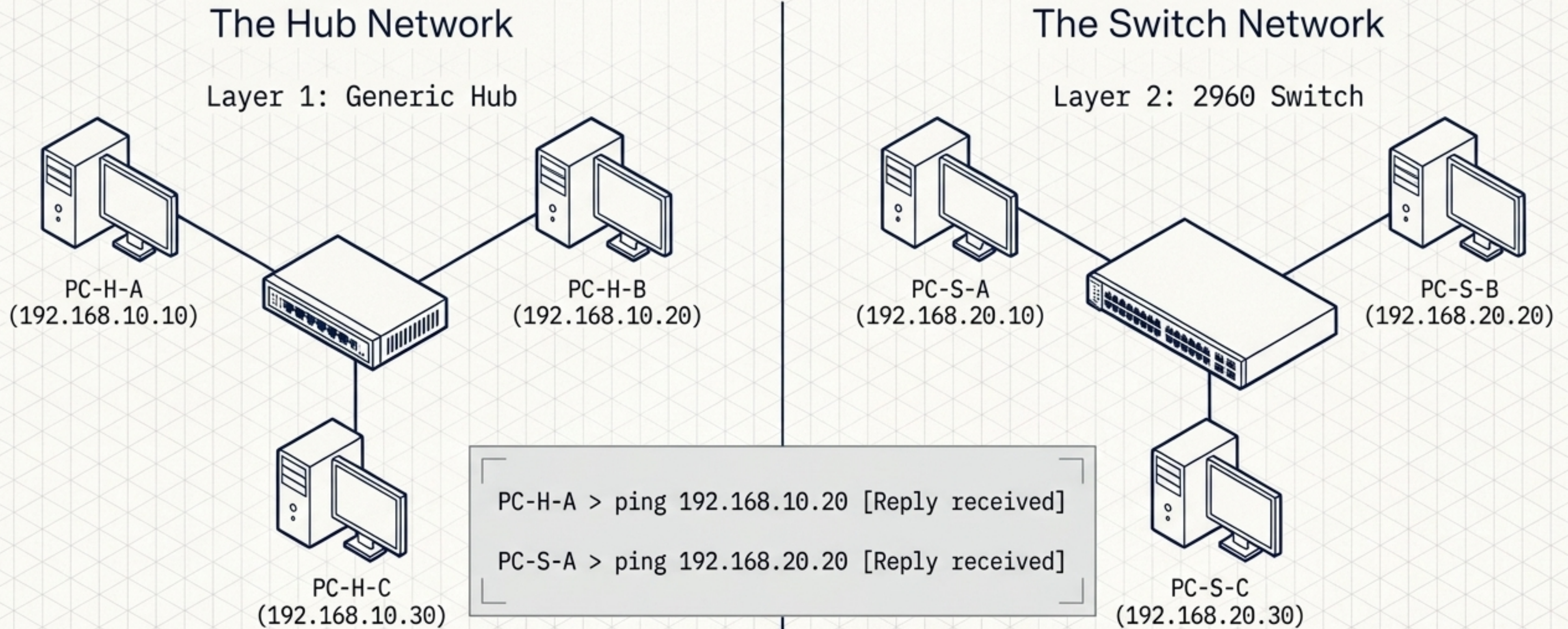
Topology C: Wireless Bridging



Topology D: Security & Gateways

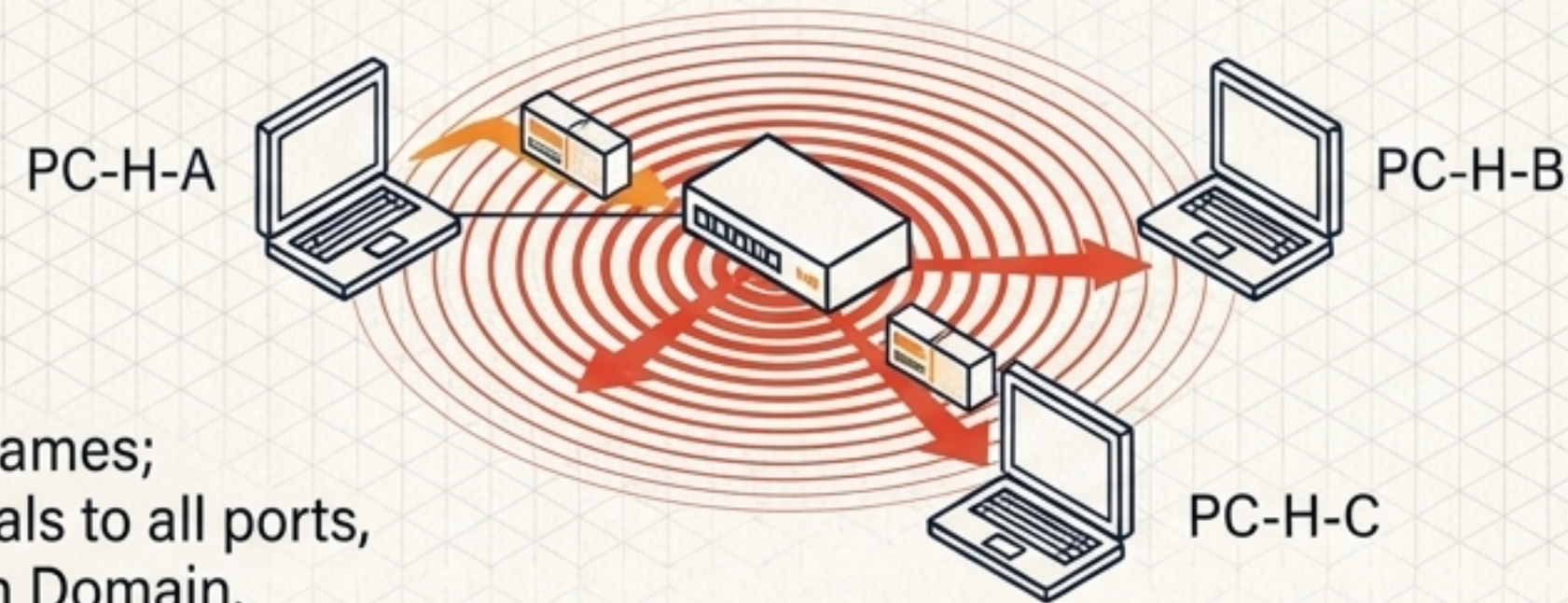


Establishing the Baseline: L1 vs L2



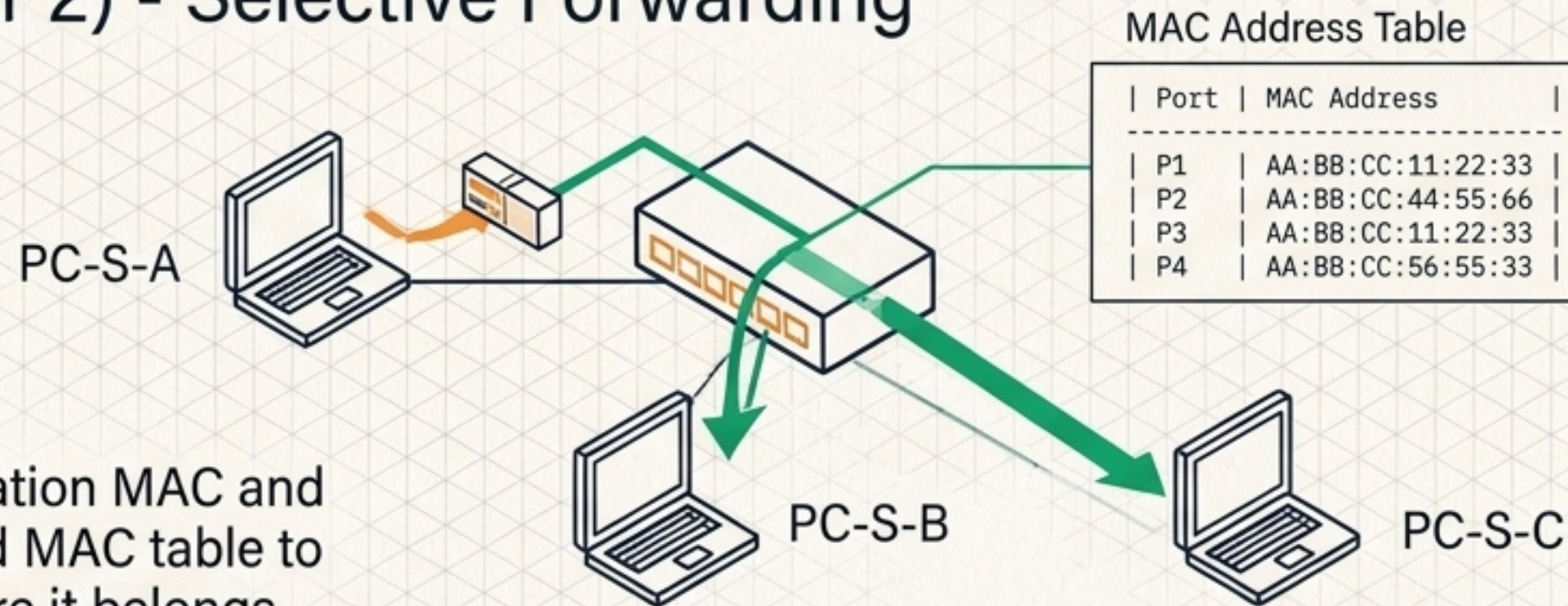
Both networks achieve basic ping connectivity. But HOW they move the data reveals the fundamental gap between Layer 1 hardware and Layer 2 intelligence.

The Hub (Layer 1) - Blind Repetition



Hubs do not read Ethernet frames; they duplicate electrical signals to all ports, creating one Shared Collision Domain.

The Switch (Layer 2) - Selective Forwarding



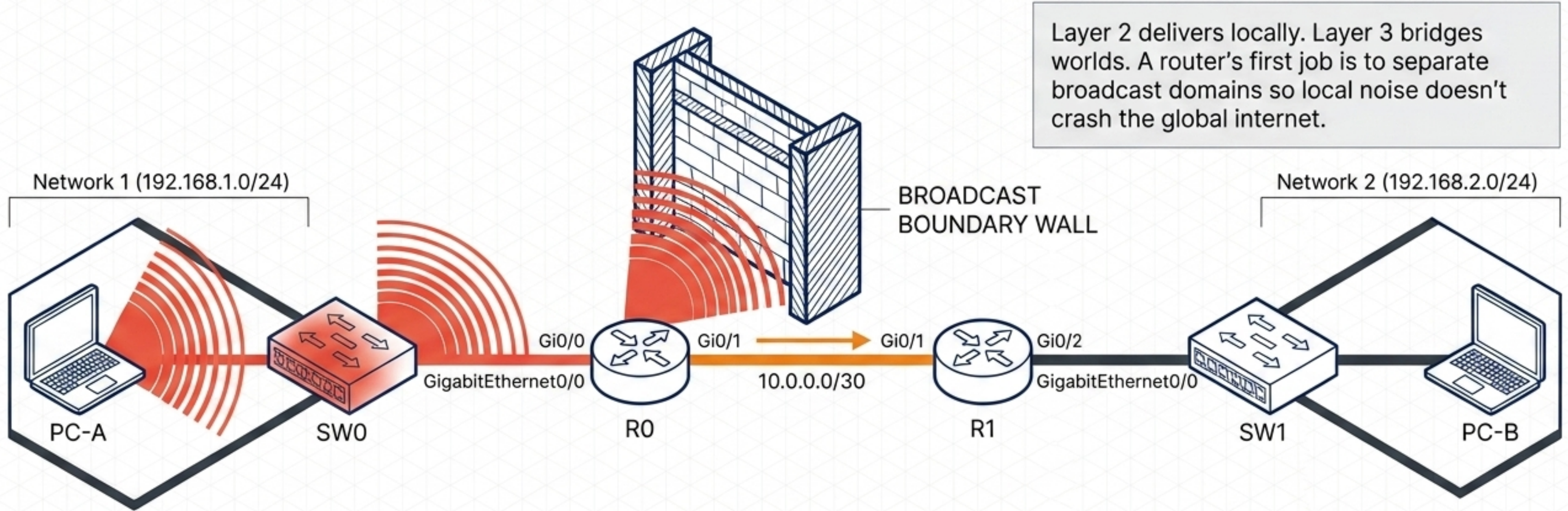
Switches read the Destination MAC and use a dynamically learned MAC table to forward data exactly where it belongs.

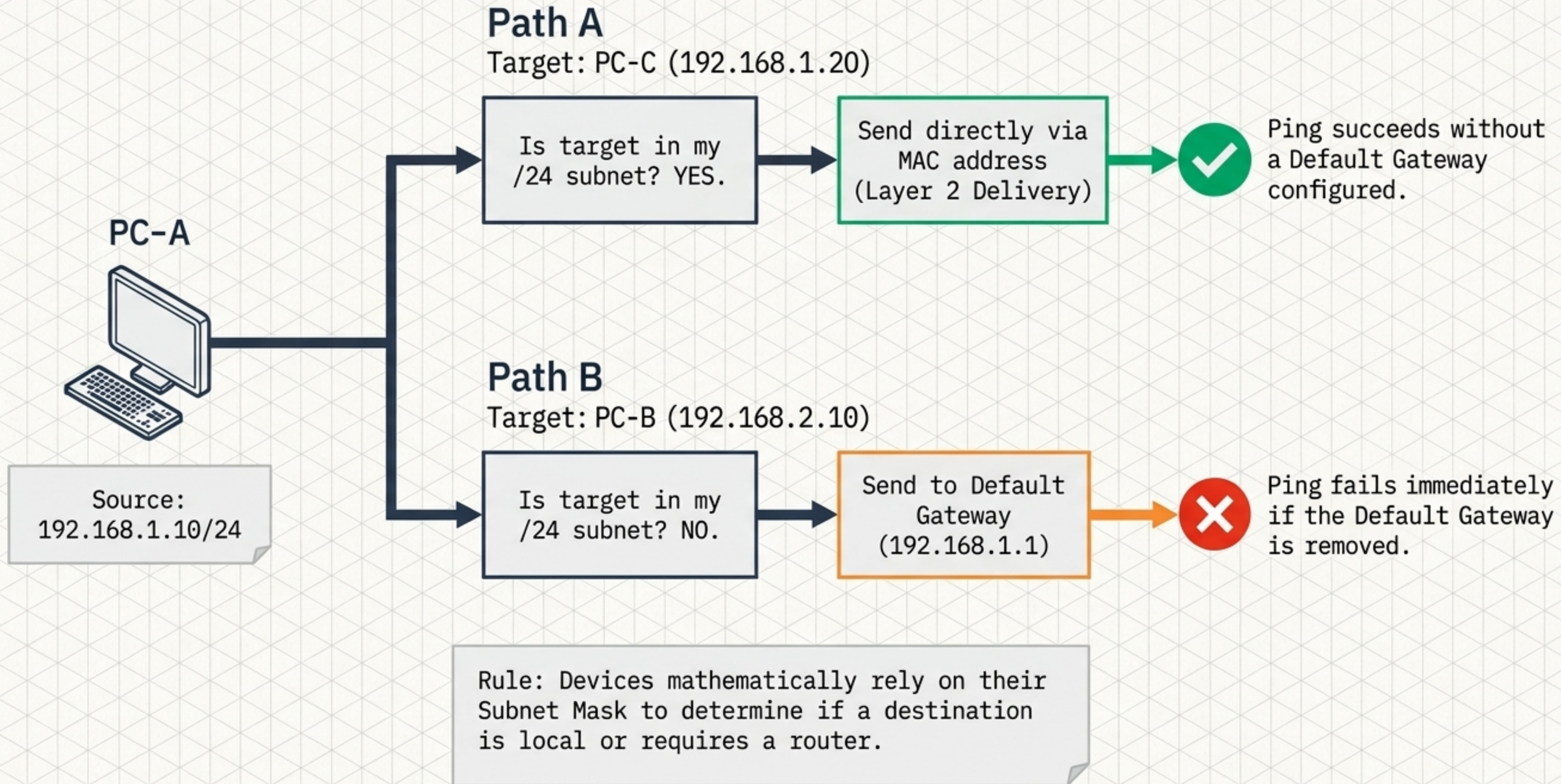
Why is this frame flooding?

Hub Traffic	Unknown Unicast (Switch)	Broadcast (Switch)
Reason: Hardware Limitation (Layer 1) Trigger: Every single frame Status: Permanent	Reason: The Switch has not learned the Destination MAC yet Trigger: Empty MAC Table Status: Temporary (until it learns the MAC from the reply)	Reason: Necessary network feature (Layer 2) Trigger: Destination MAC is explicitly FF:FF:FF:FF:FF:FF Status: By design (e.g., ARP requests)

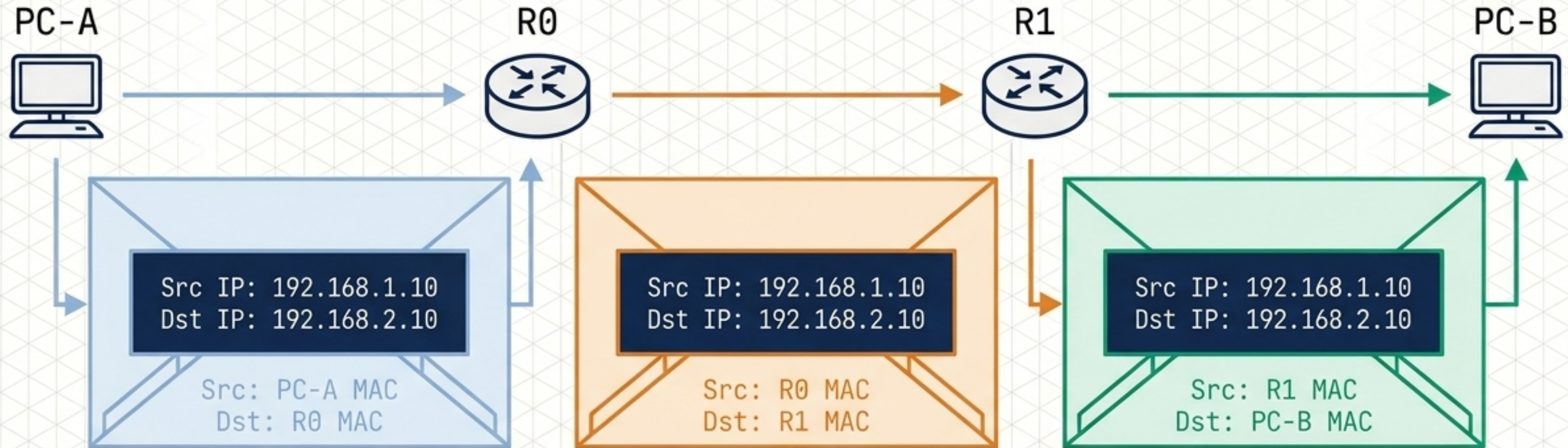
Flooding isn't always a failure. On a switch, it is either a temporary learning phase or an intentional broadcast mechanism.

Crossing the Broadcast Boundary





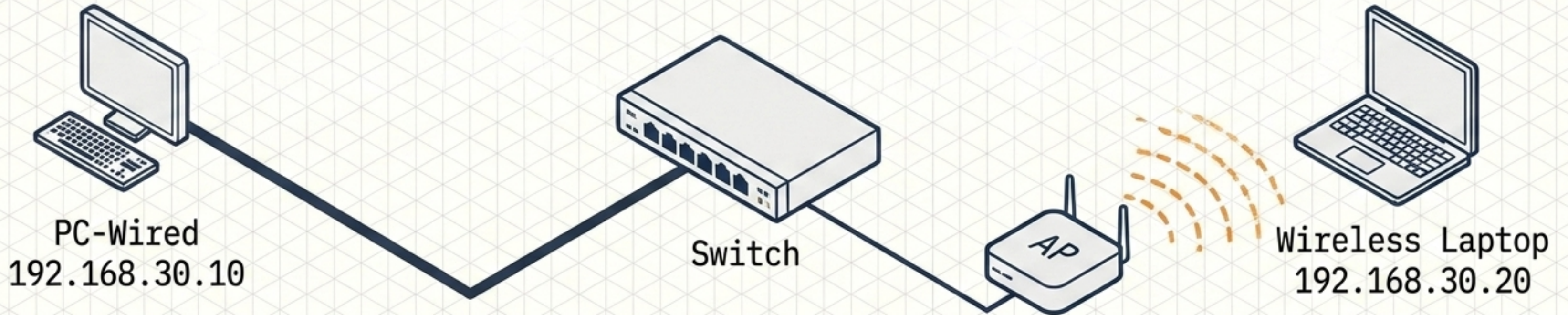
The Great MAC Rewrite (Hop-by-Hop)



IP addresses dictate the final destination. MAC addresses dictate the very next physical step. The router rebuilds the Layer 2 frame at every single hop.

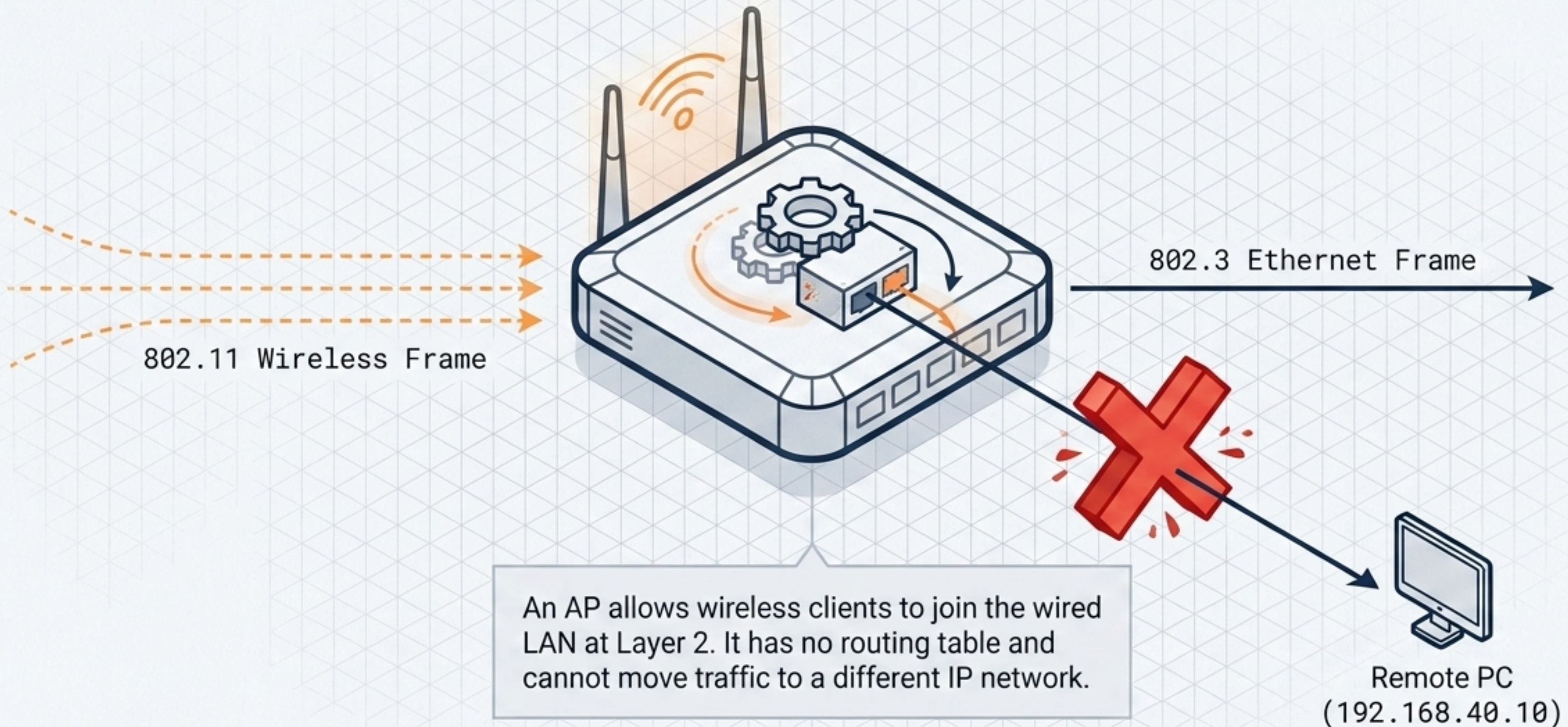
Topology C: The Wireless Bridge

```
Laptop > ping 192.168.30.10  
Success (No Router Required) ✓
```



Introducing the Access Point. Despite the shift in physical media (radio vs. copper), both devices exist on the exact same IP subnet (192.168.30.0/24).

Access Points Bridge, They Don't Route



Topology D: Security Policy (The Firewall)

Trusted LAN



PC-Trusted
10.10.10.10

R-GW



Pings succeed outbound, but fail inbound. Why? Because routing and security are distinct mechanisms operating on the same physical box.



WAN Server
10.20.20.10

```
access-list 101 deny icmp 10.20.20.0 0.0.0.255 any
```

```
interface GigabitEthernet0/1  
ip access-group 101 in
```

Routing vs. Security

The Map Reader (Routing)



Question answered: “Where should this packet go?”

Mechanism: Forwards based on Destination IP.

The Bouncer (Firewall/ACL)

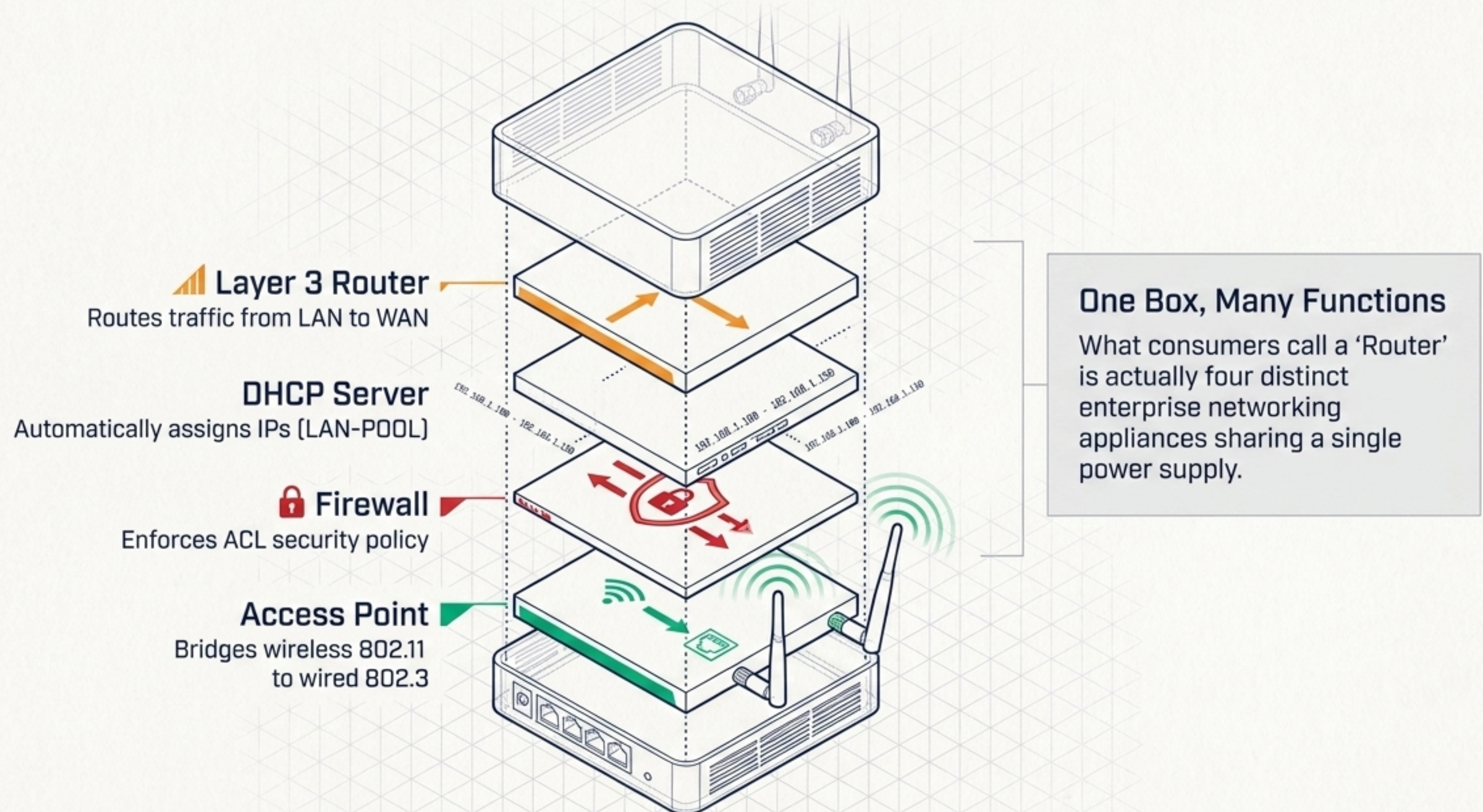


Question answered: “Is this traffic allowed?”

Mechanism: Blocks or permits based on Policy Rules.

A router fundamentally **wants to deliver data**. A firewall tells it when to say **no**.
A blocked packet does not mean the routing table is broken.

Synthesis: Exploding the Home Gateway



The Master Device-Function Code

Device	OSI Layer	Information Used	Internal Table	The Question it Answers
Hub	L1	Physical Signal	None	“Which ports to repeat to?”
Switch	L2	<u>MAC Address</u>	MAC/CAM	“Which port has this MAC?”
Router	L3	<u>IP Address</u>	Routing Table	“Which network for this IP?”
Access Point	L2	Wireless MAC	None (Bridge)	“How do wireless clients join the LAN?”
Firewall	L2-L7	Policy Rules	ACL/Ruleset	“Should this traffic be allowed?”

The Golden Rules of the Physical World

1

Same Subnet = Layer 2 Delivery.

A switch or AP is all you need to communicate locally.

2

Different Subnet = Layer 3 Routing.

A router and a default gateway are strictly required to cross boundaries.

3

Security is Independent of Routing.

A device can know exactly how to reach a destination, yet still be instructed by a Firewall to drop the packet.

4

Hardware is Logical.

One physical plastic box can simultaneously run a Router, Switch, DHCP Server, Firewall, and AP.

```
> Lab_Session_Closed.  
> Exit 0.
```